

# SOHEE PARK

Hands-on Senior AI/ML Engineer | Agentic AI & LLM Systems | Production ML

github.com/Sohee-sophiePark • linkedin.com/in/sohee-sophie-park • Toronto, ON

## PROFESSIONAL SUMMARY

Hands-on Senior AI/ML Engineer with 6 years of experience designing, building, and shipping production AI: multi-agent systems, RAG, LLM evaluation pipelines, and classical ML. I own systems end to end, from data and context pipelines through agent architecture, evaluation gates, CI/CD, and phased rollout. Recent systems I architected and built include Wealth Advice+, an AI insight platform for wealth advisors, and a multi-agent brief system serving ~3.8K advisors across a ~10M-customer base. Experienced in regulated environments where traceability, evaluation rigor, and human-in-the-loop design are requirements.

## TECHNICAL SKILLS

**Agentic AI & LLMs:** Multi-Agent Systems, Google Agent Development Kit (ADK), LangGraph, LangChain, RAG, Prompt & Context Engineering, Structured Output Enforcement, Vertex AI (Gemini), Anthropic SDK

**LLM Evaluation & MLOps:** DeepEval, Langfuse (tracing), MLflow, Prompt Regression Testing, Evaluation-Gated CI/CD, Git

**Machine Learning:** XGBoost, Bayesian Optimization, Imbalanced Classification, Churn & Transaction Prediction, NLP (sentiment, clustering)

**Languages:** Python, SQL, PySpark, JavaScript

**Cloud & Data:** GCP (Vertex AI, BigQuery, GCS, Cloud Composer), AWS, Azure, Databricks, Airflow, dbt, FastAPI, MinIO/Parquet, MS SQL Server

**Integrations & Tools:** Salesforce CRM data, IBM OpenPages, SAP, Power BI, Streamlit, JupyterHub, Jira

## PROFESSIONAL EXPERIENCE

**Senior Manager, AI Practitioner (Data Science)** · SCOTIABANK, Toronto

Dec 2025 – Present

*Technical lead and hands-on builder: I architect and code each system below and mentor a team of 6 AI/ML engineers.*

- **Wealth Advice+ AI Insights:** Architected and built Gemini-powered Vertex AI agents and PySpark/BigQuery context pipelines over Salesforce CRM data that generate wealth upsell recommendations for advisors, with business impact validated by Finance. Shipped through a Power BI delivery layer during a CRM migration; native CRM integration is underway.
- **Retail Advisor Meeting Briefs:** Designed and built a multi-agent system that generates client briefs from CRM data for ~3.8K advisors serving a ~10M retail customer base. Built the evaluation stack (DeepEval, Langfuse tracing, MLflow) that gated a district-by-district rollout.
- **Evaluation-Gated CI/CD:** Standardized CI/CD with prompt regression tests and evaluation benchmarks (DeepEval/MLflow) as required merge checks, enforced by both CI and review policy, cutting ticket-to-merge cycle time by 35%.

**Manager, Data Science & Analytics Engineering** · SCOTIABANK, Toronto

Jul 2022 – Nov 2025

- **Predictive UX:** Built the model and data pipeline (~50% of build effort) for the AI engine that analyzes clients' transaction history to pre-fill one-click prompts; defined the API contract, latency requirements, and model and adoption metrics with backend engineering. Extended it from bill payments to e-Transfers and account transfers by tuning penalty parameters in a delta-frequency model. Bill-pay prompts alone reached 260K+ clients/quarter with 92% completion.
- **AML/TM & EDD:** Built a human-in-the-loop agentic AI framework on GCP (Vertex AI, BigQuery) integrated with IBM OpenPages that pulls case data from source systems, routes it by domain (Transaction Monitoring, Enhanced Due Diligence, Suspicious Transaction Reporting), summarizes evidence, and flags risk indicators; investigators review every output and make all final decisions. Cut investigator time per case by 50%+.
- **WealthNet Hub (RAG):** Built a RAG retrieval system over Wealth Management internal documents (LangChain, Gemini, Chroma, OCR ingestion), reaching 89% top-3 retrieval accuracy (hit rate@3) on an SME-defined test set of 30+ questions benchmarked against manual search.
- **GRM Credit Committee Summarizer:** Built an LLM pipeline (Vertex AI Gemini, GCS, BigQuery, Streamlit) that condenses 25-page Senior Credit Committee submissions into 2–4 page executive briefs (~85–90% shorter) while preserving key credit components: borrower assessment, financial performance, capital structure, strategy, and repayment capacity. Delivered a business-endorsed 20–30% efficiency gain, freeing an estimated 1,000+ hours per year.
- **Mutual Fund Attrition Model:** Built an XGBoost churn model with Bayesian optimization on imbalanced data, calibrating the decision threshold to retention team capacity: the top 2 deciles captured 80% of actual churners at 4x lift over random outreach. Automated pipelines with PySpark and parallel Airflow jobs into MinIO/Parquet, later replicated on GCP with dbt, Cloud Composer, and Spark into BigQuery.

**Big Data Developer** · SDK TEK, Calgary

Dec 2021 – Jul 2022

- Built scalable ELT pipelines in Azure Databricks, cutting processing time by 85%. Implemented SCD Type 1 and Type 2 logic with automated workflows for accurate data versioning across reporting systems.

**SAP Application Developer** · IBM, Halifax

Jul 2020 – Dec 2021

- Built sentiment analysis and clustering models on customer feedback using IBM Watson APIs and Python NLP. Designed SAP applications and Open SQL scripts for Public Sector cloud projects with Azure-based portal integration.

## CERTIFICATIONS

AWS Certified Machine Learning – Specialty • Microsoft Certified: Azure Data Engineer Associate • AWS Certified Developer – Associate